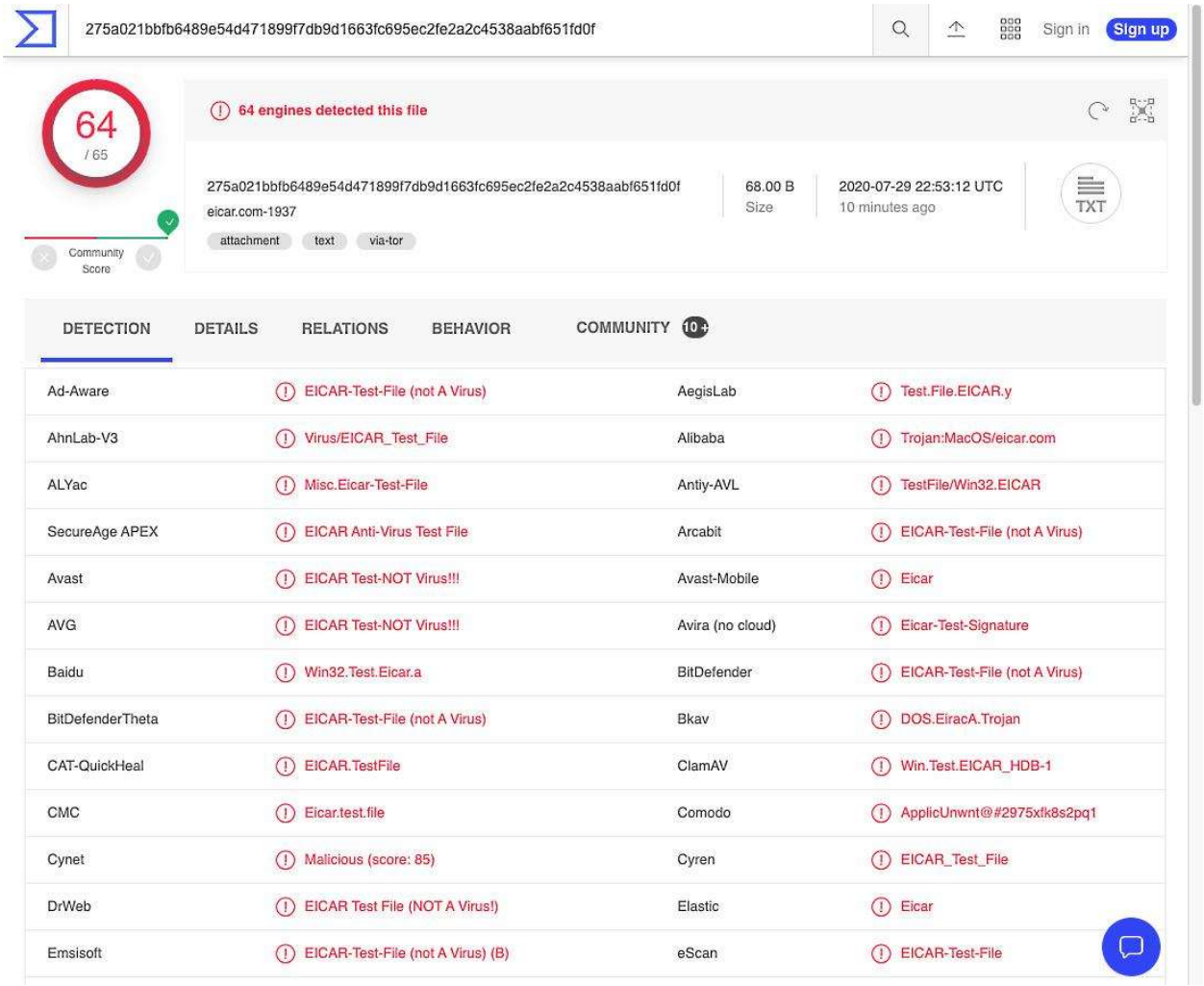
1.Create a table listing the main differences between viruses, worms, ransomware, Trojans, spyware, and adware, and give a real-world app or website example where each type of malware could potentially cause harm.

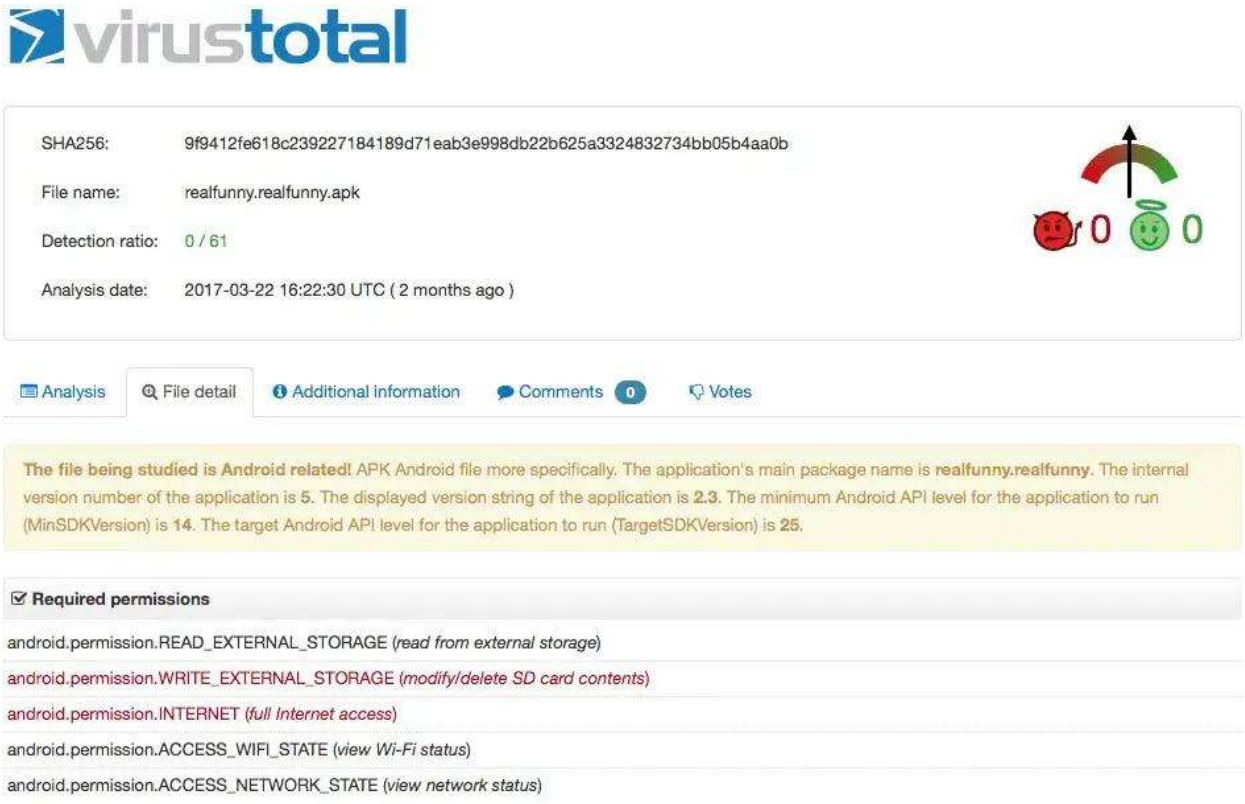
Malware Type	Meaning	Example of Possible Harm
Virus	Attaches itself to a file and spreads when the infected file runs.	A downloaded file from an unsafe website may infect a computer and damage files.
Worm	Spreads automatically from one computer to another.	A company network may be infected, affecting many computers.
Ransomware	Locks or encrypts files and demands money to restore access.	A business may lose access to its important files and customer records.
Trojan	Looks like a useful app but secretly performs harmful actions.	A fake Android app may steal login details.
Spyware	Secretly collects personal information from a device.	A fake shopping app may steal passwords or personal data.
Adware	Shows unwanted advertisements, sometimes tracking user activity.	A suspicious website or app may show pop-up ads and redirect users to unsafe websites.

2.Download a harmless sample EICAR test file (not an actual virus) and upload it to VirusTotal. Take a screenshot of the detection results and write a short summary explaining which antivirus engines detected it and what labels they gave.



**3.Pick any free Android APK from a third-party site (not Google Play), upload it to VirusTotal, and analyze the report. List at least three suspicious indicators VirusTotal highlights, such as risky permissions or flagged behaviors.

Hint: Focus on the 'Details' and 'Behavior' tabs in the VirusTotal report.**



**4.Watch an online video demo (YouTube) of a ransomware attack in a sandboxed environment. Write a step-by-step summary of what happens to the victim's files and system, and explain how the sandbox prevents real damage.

Hint: Search for 'ransomware demo sandbox' on YouTube.**

Steps to observe in a demo:

1. The user runs a ransomware sample inside a sandbox.
2. The ransomware starts running in the test environment.
3. It searches for files, such as documents, images, and other data.
4. It encrypts the files, making them difficult to open.
5. A ransom note appears, demanding money to recover the files.
6. The security researcher observes the changes and records the results.

5.Imagine you are building a Zomato-style food delivery app. List two ways malware (like spyware or adware) could affect your users, and suggest one security feature you would add to protect them.

Two ways malware could affect users:

1. Spyware: It could steal users' login details, personal information, or payment-related information.
2. Adware: It could show unwanted advertisements, redirect users to unsafe websites, or disturb the ordering process.

One security feature:

I would add multi-factor authentication (MFA). It requires an additional verification step when a user logs in or performs a sensitive action.